

EV Charger Data Integration & Augmentation

- COMP5339 Project Assignment 1 -

In this data engineering assignment, your team will acquire and integrate electric vehicle (EV) charger data from Transport for NSW with spatial data from the Australian Bureau of Statistics (ABS), and augment it with charger details obtained from EV charger operator websites. The project aims to provide practical experience in core data engineering skills such as data retrieval, integration, cleaning, transformation, and storage.

This is a **group assignment**, comprising teams of 3 to 4 students. **The groups formed for Assignment 1 will remain unchanged for Assignment 2**, which will build on the work completed in Assignment 1.

These two assignments represent two stages with separate submissions:

- **Stage 1 (Week 8):** Submit all Assignment 1 deliverables by **23:59, Friday 25 Sep 2026**.
- **Stage 2 (Week 12):** the instructions for Assignment 2 will be provided later.

All submissions will undergo plagiarism checks.

Assignment 1 Tasks

This Assignment 1 concentrates on acquiring and integrating static datasets about electric vehicle (EV) chargers in NSW for coverage analysis – and in Assignment 2 for price monitoring. Your team needs to develop Python programs to complete the following tasks:

1. Data Acquisition

Programmatically retrieve the following datasets (using Python or Unix scripts):

- the locations of EV chargers in NSW from either the Transport for NSW website (<https://opendata.transport.nsw.gov.au/data/dataset/ev-charging-locations>) or via data.gov.au (<https://data.gov.au/data/dataset/nsw-2-ev-charging-locations>). Note that for the Transport for NSW website, you need to create a (free) login. Retrieve the most-recent version from December 2025;
- digital boundary shapefile on the SA4 level (regional) from the latest release of the Australian Statistical Geography Standard (ASGS) from the ABS (<https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs/edition-4-july-2026-june-2031/access-and-downloads/digital-boundary-files>).

Your solution should be fully reproducible. Dataset retrieval should be automated through scripts or Python and should not require manual downloading.

2. Data Integration and Cleaning

Write Python code to pre-process, clean, and combine the retrieved data. This may include converting data types, and identifying and addressing data quality issues such as missing values, duplicate records, inconsistent naming conventions and inconsistent charger attributes. As part of this process, you should also spatially join the EV charger and ASGS data (with Python or SQL) so that you add an additional field to the EV chargers that represents the SA4 region in which it is located.

3. Data Augmentation

Augment a substantial portion (target: 50% or more) of EV fast charger locations (charger type DC) with one or more additional attributes not present in the original Transport for NSW dataset. Examples include:

- Plug types,
- Pricing information,
- Operator details,
- Number of charging bays.

You can concentrate here on the DC chargers. Perform this programmatically (using Python) by querying the corresponding information from either the EV charger operator's website, or another open Web API such as Open Charge Map or even Google Maps. Document methods and API usage. Document any matching strategies used to link external data to charger locations (e.g. name matching, coordinate matching, address matching). Keep a local copy of the retrieved data (e.g. in CSV format), as repeated retrieval may be time-consuming and subject to API limits.

4. Data Transformation and Storage

Transform the processed and augmented data into a structured format suitable for analysis and visualization. Specifically, you should

- design a relational database schema that supports efficient storage and querying of charger locations, operators, charger characteristics, and geographic regions; and
- implement this schema and store your data (programmatically) in DuckDB. Make sure that you include the spatial extensions so that we can run some spatial queries in Assignment 2 (also helpful for Task 2).
- Provide SQL DDL scripts that recreate your schema.

Important Note: Clearly justify your database design decisions (e.g., normalized or denormalized schema) in your project report. If your group encounters significant difficulties working with a database, you may alternatively store your data in separate CSV files; however, choosing CSV storage will result in a mark penalty.

5. Documentation and Reporting

Maintain detailed documentation throughout your project workflow. Your final report should clearly summarise your methods for data retrieval, integration, cleaning, augmentation, and transformation/storage. Specifically:

- Explain your data retrieval and preprocessing methods, including the spatial integration between EV chargers and ASGS regions.
- Identify additional web APIs or web sources which you used for data augmentation, and describe how you accessed these APIs and/or web sources.
- Provide a clear diagram of your database schema and justify your design choices.
- Discuss the completeness and consistency of both original and augmented datasets.
- Describe challenges encountered and how you overcame them.
- [Recommended report sections on next page]

- **Recommended Report Sections**
 - Introduction
 - Data Sources and Retrieval
 - Data Cleaning and Quality Assessment
 - Spatial Data Integration Methodology
 - Data Augmentation Methodology
 - Database Design
 - Challenges and Limitations
 - Team Contributions

Deliverables

Submit the following three items via Canvas:

1. Source Code and Database Package (.zip)

containing scripts, Python files (.py or .ipynb), data, **requirements.txt** and **README.md**

- Organise your code and Python notebooks in a clear and logical project structure.
- Include a **requirements.txt** file listing all Python packages required to run your program within a clean Python virtual environment.
- Include your **final database** from Task 4 with
 - a **DDL script**; and
 - the DuckDB database file.
 (groups choosing the CSV fallback should submit the processed CSV files instead)
 - You may submit additional data obtained during **Data Augmentation** as **CSV** files.
- Include a **README.md** file (using Markdown format) containing:
 - project structure,
 - setup instructions,
 - execution instructions,
 - assumptions made,
 - description of outputs produced and any included data files.

2. Project Report (.pdf)

- Provide a clearly structured report of **up to 6 pages** of content. The title page and any optional appendix do not count towards the page limit. The title page must show your group name and list the team members by name and SID. See Task 5 above for recommended report sections.
- Your report must include a short paragraph not exceeding 100 words, detailing the contribution of individual team members. For example:
 - AXX: Led data collection and preprocessing, ...
 - BXX: Developed the database model ...

Submission Deadline

All Assignment 1 deliverables are due by **23:59, Friday 25 September 2026**.
Late submissions will incur penalties according to university policy.